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**Definition. Binomial Distribution.** A discrete random variable X is said to have a binomial distribution if its probability function is defined by

$$p(x) = {}^nC_x p^x q^{n-x} \quad ; x = 0, 1, 2, \ldots\ldots\ldots, n.$$

where $p + q = 1$.

This distribution is called binomial distribution, since the probability for different values of X are the various terms of the binomial expansion.

$$(q+p)^n = q^n + {}^nC_1 p^x q^{n-1} + {}^nC_2 p^x q^{n-2} + \ldots\ldots\ldots + p^n.$$

Here p is the probability of success in a single Bernoulli trial and q is the probability of failure. It is to be noted that n and p are the two unknown constants of the distribution. They are known as the parameters of the distribution. Here n is the number of Bernoulli trials and p is the probability of successes.

The binomial variable X with parameter n and p is symbolically written as $X \sim B(n, p)$

***Parameter***

The unknown constants required to define a distribution are called the parameters of the distribution. A distribution is completely known if the parameters are known. It is to be noted that any numerical value calculated from the population is also called parameter.

**10.3.1. Mean and Variance of the distribution.** If n and p are two parameters of the binomial distribution, then mean and variance of the distribution are

$$\text{Mean} = \mu = E[X] = np \quad \text{and} \quad \text{Variance} = \sigma^2 = npq.$$

$$\text{Standard deviation} = \sigma = \sqrt{npq}.$$

It is to be noted that mean is greater than variance of the distribution since $q < 1$.

**10.3.2. Some important properties of the distribution**

1. It is a discrete probability distribution with parameters n and p.
2. The mean of the distribution is np and its variance is npq. The mean of the distribution is greater than variance since $q < 1$.
3. The distribution is positively skewed if $p < 1/2$ and negatively skewed if $p > 1/2$
4. The distribution is symmetric if $p = q = 1/2$.
5. The distribution tends to Poisson distribution if the number of trials, n tends to infinity.



6. The distribution tends to normal distribution if n tends to infinity and p or q is not so small.
7. $P(X = n) = p^n$ and $P(X = 0) = q^n$

**10.3.3. Some practical applications of Binomial distribution**

1. Number of defective items in a randomly selected sample of 12 products.
2. Number of days of increasing price index in share market in a randomly selected 15 days.
3. Number of correct answers in a multiple choice test if a student answers all the questions randomly.
4. Number of workers suffer from occupational disease in a randomly selected sample of 10 workers.
5. Number of female babies in society.
6. Number of successful hits in a target out of fixed number of hits.
7. Number of customers buy a particular commodity.

**Example 10.3.1.** Suppose X is a binomial variate with parameters $n = 4$ and $p = 0.45$. Find the following probabilities

(i) $P[X = 2]$; (ii) $P[X > 3]$ and (iii) $P[X \le 2]$.

**Solution.** We have $n = 4$ and $p = 0.45$, then $q = 1 - p = 1 - 0.45 = 55$, i.e. $X \sim B(4, 0.45)$.

Then the probability function of the binomial variable X is

$$P[X = x] = p(x) = {}^nC_x p^x q^{n-x} \quad x = 0, 1, 2, 3, 4$$

$$= {}^4C_x \, (0.45)^x (0.55)^{4-x}; x = 0, 1, 2, 3, 4$$

(i) $P[X = 2] = p(2) = {}^4C_2 \, p^2 q^{4-2}$

$$= {}^4C_2 \, (0.45)^2 (0.55)^2 = 10(0.2025)(0.3025) = 0.3675$$

(ii) $P[X > 3] = P[X = 4] = p(4) = (0.45)^4 = 0.0410$

(iii) $P[X \le 2] = P[X = 0] + P[X = 1] + P[X = 2] = p(0) + p(1) + p(2)$

$$= 0.0915 + 0.2995 + 0.3675 = 0.7585.$$

**Matched problem**

**Problem.** For a binomial distribution with $n = 7$ and $p = 0.2$, find

(i) $P[X = 5]$; (ii) $P[X < 8]$ and (iii) $P[X \ge 4]$

Ans. (i) 0.0043; (ii) 1 (iii) 0.0333.

**Example 10.3.2.** The probability that a patient recovers from a rare blood disease is 0.4. If 5 people are known to have contracted this disease, what is

the probability that (i) exactly 3 survive,(ii) at least two survive (iii) at most two survive and (iv) none survive.

**Solution.** Here, we define survive as success and not survive as failure. Let X be the number of people that survive. Then X is a binomial variate with probability of success p = 0.4 and probability of failure q = 1 - p = 1 - 0.4 = 0.6 and n = 5, i.e, . X ~ B(5, 0.4). Then according to binomial distribution, the probability function of X is

$$P[X = x] = p(x) = {}^nC_x p^x q^{n-x} \; ; \quad x = 0, 1, 2, \ldots\ldots, n.$$

$$= {}^5C_x (0.4)^x (0.6)^{5-x} ; \quad x = 0, 1, 2, 3\ 4, 5.$$

The probabilities for different values of X are

$P[X = 0] = p(0) = (0.6)^5 = 0.07776.$

$P[X = 1] = p(1) = {}^5C_1(0.4)(0.6)^4 = \frac{5!}{4!1!}(0.4)(0.1296) = 5\times0.0518 = 0.2592.$

$P[X = 2] = p(2) = {}^5C_2(0.4)^2(0.6)^3 = \frac{5!}{2!3!}(0.16)(0.216) = 10\times0.03456 = 0.3456.$

$P[X = 3] = p(3) = {}^5C_3(0.4)^3(0.6)^{5-3} = \frac{5!}{3!2!}(0.4)^3(0.6)^2 = 10\times.064\times0.36 = 0.2304.$

$P[X = 4] = p(4) = {}^5C_4(0.4)^4(0.6) = \frac{5!}{4!1!}(0.0.0256)(0.6) = 5\times0.01536 = 0.0768.$

$P[X = 5] = (0.4)^5 = 0.01024.$

(i) P [Exactly 3 survive] = P [X = 3] = 0.2304.

(ii) P [at least 2 survive] = P [X ≥ 2]

$= P[X = 2] + P[X = 3] + P[X = 4] + P[X = 5]$

$= 0.3456 + 0.2304 + 0.0768 + 0.01024 = 0.66304.$

(iii) $P[X \le 2] = P[X = 2] + P[X = 1] + P[X = 0]$

$= 0.3456 + 0.2592 + 0.07776 = 0.65376.$

(iv) $P[X = 0] = p(0) = 0.07776.$

MBA

**Example 10.3.3.** The probability that a MBA graduate from a private University gets an executive job is 0.6. Four MBA graduates applied for executive jobs. What is the probability that (i) exactly 2 will get the job, (ii) all will get the job, (ii) at least 3 will get the job.

**Solution.** We define a MBA gets a job as success and does not get a job as failure. Let X be the number of MBA graduates get the job. Then X is a binomial variate with p = 0.6, q = 0.4 and n = 4, i,e, X ~ B(4, 0.6).



Then the probability function of X is

$$P[X = x] = p(x) = {}^nC_x p^x q^{n-x} \; ; \quad x = 0, 1, 2, \ldots\ldots, n.$$

$$= {}^4C_x (0.6)^x (0.4)^{4-x} ; \quad x = 0, 1, 2, 3, 4.$$

The probabilities for different values of X are

$P[X = 0] = p(0) = (0.4)^4 = 0.0256.$

$P[X = 1] = p(1) = {}^4C_1(0.4)^3(0.6) = \frac{4!}{3!1!}(0.4)^3(0.6) = 4\times0384 = 0.1536.$

$P[X = 2] = p(2) = {}^4C_2(0.6)^2(0.4)^2 = \frac{4!}{2!2!}(0.36)(0.16) = 6\times0.0576 = 0.3456.$

$P[X = 3] = p(3) = {}^4C_3(0.6)^3(0.4) = \frac{4!}{3!1!}(0.6)^3(0.4) = 4\times.0.216\times0.4 = 0.3456.$

$P[X = 4] = p(4) = (0.6)^4 = 0.1296.$

(i) P [X=2] = 0.3456.

(ii) P [X= 4] = 0.1296.

(iii) $P[X \ge 3] = P[X=3] + P[X=4] = 0.3456 + 0.1296 = 0.4752.$

**Example 10.3.4.** Suppose X is a binomial variate with mean 4 and variance 3. Find the probability distribution of X.

**Solution.** Let n and p be the two parameters of the distribution. Here, we have

Mean = np = 4 and variance = npq = 3

Hence, $q = \frac{npq}{np} = \frac{3}{4}$. The $p = 1 - \frac{3}{4} = \frac{1}{4}$.

n = 4/p = 4 × 4 = 16.

Hence the required binomial distribution is

$$f(x; n, p) = f(x; 16, \tfrac{1}{4}) = \binom{16}{x}\left(\frac{1}{4}\right)^x\left(\frac{3}{4}\right)^{16-x} ; \quad x = 0, 1, 2, \ldots\ldots, 16$$

**Matched problem**

**Problem.** Suppose X is binomial variate with mean 9 and variance 3.6. Find the binomial distribution. Ans p = 3/4, n = 15

$$f(x; n, p) = f(x; 15, \tfrac{3}{4}) = \binom{15}{x}\left(\frac{3}{4}\right)^x\left(\frac{1}{4}\right)^{15-x} ; \quad x = 0, 1, 2, \ldots\ldots, 15.$$

**Exercise 10.3.5.** Warranty records show that the probability that a new car needs a warranty repair in the first 90 days is 0.05. If a sample of three new cars is selected, what are the probabilities that in the first 90 days

(i) None needs a warranty repair?
(ii) At least one needs a warranty repair?
(iii) More than one needs a warranty repair?
(iv) What are the mean and standard deviation of the number of cars need warranty repair.

**Solution.** Here X is the number of cars need warranty repair. X is a binomial variate with parameters n = 3 and p = 0.05. Then q = 1-0.05 = 0.95. Then the probability function of X is

$$P[X = x] = p(x) = {}^nC_x p^x q^{n-x} \; ; \qquad x = 0, 1, 2, \ldots\ldots, n.$$

$$= {}^3C_x (0.05)^x (0.95)^{3-x} ; \quad x = 0, 1, 2, 3.$$

(i) The probability that non-needs warranty repair is

$$P[X = 0] = p(0) = (0.95)^3 = 0.8574$$

(ii) The probability that at least one needs repair is

$$P[X>0] = 1 - P[X = 0] = 1 - 0.8574 = 0.1426.$$

(iii) The probability that more than one needs repair is

$$P[X>1] = 1 - P[X \le 1] = 1 - (P[X = 0] + P[X=1])$$

$$= 1 - (0.8574 + 0.1354) = 0.0072.$$

(iv) The mean of the distribution is $\mu = np = 3 \times 0.05 = 0.15$.

The standard deviation of the distribution is

$$\sigma = \sqrt{npq} = \sqrt{3 \times 0.05 \times 0.95} = 0.38.$$

**Example 10.3.6.** Admission tests in different faculties of Chittagong University are held in MCQ method. There are ten MCQs in Bangla, each carrying one mark, are provided for each subject. There are five options in each question only one of which is correct answer. Suppose, a candidate has appeared in the test in a faculty without any preparation in Bangla. S(he) has to answer all the questions randomly (i) Write down the distribution of correct answers, (ii) find the probability that only three of such answers are correct, (iii) if the pass mark for that subject is 5, find the probability of passing in the subject.

**Solution:** (i) Let X be the number of correct answers. Here, the number of correct answers can take the values 0, 1, ....., 10, and one answer out of 5 is correct, so the probability of a correct answer is p = 1/5 = 0.20, the probability of an incorrect answer is q = 1 – p = 0.80, which remain constant



from question to question. Thus, X follows binomial distribution with n = 10 and p = 0.20, i, e, X ~ B(10, 0.20). The probability function of X is given by

$$P[X = x] = {}^nC_x p^x q^{n-x} \; ; x = 0, 1, 2, \ldots\ldots 10$$

$$\text{Or, } P[X = x] = {}^{10}C_x (0.2)^x (0.8)^{10-x} \; ; x = 0, 1, 2, \ldots\ldots 10$$

(ii) $P[X = 3] = {}^{10}C_3 (0.20)^3 (0.80)^{10-3} = 120(0.20)^3 (0.80)^7 = 0.20113$

(iii) To pass in that subject, the candidate must answer at least 5 questions correctly, so the required probability is

$$P[X \ge 5] = P[X = 5] + P[X = 6] + \ldots\ldots\ldots + P[X = 10]$$

$$= 0.026424 + 0.005505 + \ldots\ldots\ldots + 0.000000102 = 0.03793$$

## 10.4. Poisson Distribution

Poisson distribution was discovered by the French mathematician and physicist Simeon Denis Poisson [1781-1840], who published it in 1837. It has many applications in modern day life such as modeling the radioactive decay, predicting the number of telephone calls arriving at an exchange, etc. Poisson distribution is a limiting case of the binomial distribution under the following conditions:

(i) The probability of success or failure in Bernoulli trial is very small. That is $p \to 0$ or $q \to 0$,
(ii) n, the number of trials is very large and
(iii) $np = \lambda$ (say) is a finite constant.

**Definition. Poisson distribution.** A discrete random variable X is said to have a Poisson distribution if its probability function is defined by

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!} \; ; x = 0, 1, 2, \ldots\ldots\ldots\ldots, \infty$$

where e = 2.71828 and $\lambda$ is the only parameter of the distribution which is the mean of the distribution.

It can be easily shown that

i) $f(x; \lambda) \ge 0$ for each x
ii) $\Sigma f(x; \lambda) = 1$.

It can be shown that the both mean and variance of the distribution are equal to $\lambda$. The distribution is positively skewed and leptokurtic. But if the mean of the distribution tends to infinity, Poisson distribution tends to normal distribution.

The Poisson variable X with parameter $\lambda$ is written as $X \sim Po(\lambda)$

**10.4.1. Some important properties of the distribution.**

1) The mean of the distribution is $\lambda$
2) The variance of the distribution is also $\lambda$. So the mean and variance of the distribution are equal.
3) When $\lambda$ tends to $\infty$, Poisson distribution tends to normal distribution.
4) The distribution is positively skewed and leptokurtic.

**10.4.2. Conditions under which a Poisson distribution is likely to exist**

1) The events occur independently of each other
2) The events occur singly and at random in continuous space or time.
3) The events occur at a constant rate in the sense that the mean number of occurrences in the given space or interval of time is proportional to the length of space or time. For example, if the average number of cars passing through a certain street in time t is $\lambda$, then average number of cars passing through that street in time 2t is $2\lambda$, if the average number of printing mistakes per page of a book is m, then the average number of printing mistakes in every 3-page of that book is 3m, and so on.

**10.4.3. Some practical applications of Poisson distribution**

1) The number of cars passing through a certain street in time t.
2) The number of suicides reported per day of a certain area.
3) The number of faulty blades in a packet of 100.
4) The number of printing mistakes per page of a book.
5) The number of telephone calls received at a particular telephone exchange in some unit time.
6) The number of defective materials in packing manufactured by a goods concern.
7) The number of letters lost in a mail per day in a certain city.
8) The number of robbers caught per day in a certain city
9) Number of deaths from a rare disease in a locality.
10) The number of customers arriving in a departmental store per minute.
11) The number of accidents occurring in a factory over a month

**Example 10.4.1.** Given $\lambda = 4.2$, for a Poisson distribution, find (i) $P[X \le 2]$; (ii) $P[X \ge 5]$; $P[x = 8]$.

**Solution.** Here $\lambda = 4.2$, then $e^{-4.2} = 0.0150$

(i) $P[X \le 2] = p(0) + p(1) + p(2) = \frac{(4.2)^0 e^{-4.2}}{0!} + \frac{(4.2)^1 e^{-4.2}}{1!} + \frac{(4.2)^2 e^{-4.2}}{2!}$

$= 0.0150 + 0.0630 + 0.1323 = 0.0.2103.$

(ii) $P[X \ge 5] = 1 - P[X \le 4] = 1 - p(4) - p(3) - p[X \le 2]$



$= 1 - \frac{(4.2)^4 e^{-4.2}}{4!} - \frac{(4.2)^3 e^{-3}}{3!} - 0.2103$

$= 1 - 0.1844 - 0.1852 - 0.2103 = 0.4101.$

(iii) $P[X = 8] = \frac{(4.2)^8 e^{-4.2}}{8!} = 0.0360.$

**10.4.4. Poisson approximation to Binomial.** When n is large (n > 29) and p is small such that $\lambda = np < 7$, then binomial distribution can be approximated by a Poisson distribution.

MBA

**Example 10.4.2.** Suppose a life insurance company insures the lives of 5000 men aged 42. If actual studies show the probability that any 42-year old man will die in a given year to be .001, find the exact probability that the company will have to pay x = 4 claims during a given year.

**Solution.** The exact probability is given by binomial distribution as

$$P(x; .001, 5000) = \binom{5000}{4}(.001)^4(.999)^{4996}$$

For which binomial table are not available. Here np = (5000)(.001) = 5 < 7 and n is very large. So we can safely use Poisson distribution for finding the probability.

$$P[X = 4] = p(4)\ \frac{e^{-5}5^4}{4!} = \frac{(.006738)(625)}{24} = 0.175.$$

**Example 10.4.3.** For a binomial distribution with n = 30 and p = 0.04, Find the following probabilities by using Poisson approximation to the binomial: (i) P (x = 25); (ii) P [x = 3] and (iii) P [x = 5].

**Solution.** For binomial n =30 and p = 0.04

Since n =30, we can use Poisson approximation to the binomial. For Poisson

$$\lambda = n = 30 \times 0.04 = 1.2;\ e^{-1.2} = 0.30119.$$

Here $\lambda < 7$ and n is 30. So we can safely use Poisson approximation to binomial.

(i) $P[X=25] = \frac{(1.2)^{25} e^{-1.2}}{25!} = 0.0000.$

(ii) $P[X=3] = p(3)\ \frac{(1.2)^3 e^{-1.2}}{3!} = 0.0867.$

(iii) $P[X=5] = p(5) = \frac{(1.2)^5 e^{-1.2}}{5!} = 0.0062.$

**Example 10.4.4.** Suppose that the number of emergency patients X in a given day at a certain hospital follows Poisson distribution with parameter $\lambda = 20$. What is the probability that in a given day there will be (a) 15 emergency patients, (b) at least 3 emergency patients and (c) more than 20 but less than 25 patients.

**Solution.** (a) Here, we have $\lambda = 20$ and $x = 15$.

Since X follows Poisson distribution, then

$$P[X=15]=p(15)=\frac{(20)^{15}e^{-20}}{20!}=0.0516.$$

(From the table of Poisson distribution)

(b) P [at least 3 patients] = $P[X \geq 3] = 1 - P[X \leq 2]$

$$=1-p(0)-p(1)-p(2)=1-0.0000-0.0000-0.0000=1.$$

(c) $P[20 < X < 25] = p(21) + p(22) + p(23) + p(24)$

$$= 0.0846 + 0.0769 + 0.0669 + 0.0537$$

$$= 0.2441.$$

**Example 10.4.5.** In a production process, on an average, one in 400 items is defective. Suppose the items are packed in boxes of 100.

a) A box is selected from a day's output, assuming appropriate probability model find the probability that the box will contain

(i) no defective;
(ii) less than two defectives
(iii) one or more defectives

b) If the items are packed in boxes of 500, what is the probability that a randomly selected box will contain no defective?

**Solution.** (a) Here, number of defective items in a box is a random variable.

Let X be the number of defective items in a box,

The probability that an item will be defective is

$p = 1/400$, probability is very low.

$n = 100$, number of items packed in the box is quite large.

Hence the appropriate probability model is Poisson with mean,

$$\lambda = np = 100 \times \frac{1}{400} = 0.25;$$

which is the average number of defective items in a box of 100.

Thus, $X \sim Po(0.25)$ and $P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$, so,



(i) Probability of no defective

$$= P[X=0] = p(0) = e^{-\lambda} = e^{-0.25} = 0.7788$$

(ii) Probability of less than two defectives

$$= P[X \leq 1] = P[X=0] + P[X=1] = p(0) + p(1)$$

$$= e^{-\lambda} + \lambda e^{-\lambda} = e^{-0.25}(1+\lambda) = 0.7788\,(1+0.25) = 0.9735.$$

(iii) Probability of one or more defectives

$$= P[X \geq 1] = 1 - P[X=0] = 1 - p(0) = 1 - e^{-\lambda} = 1 - e^{0.25}$$

$$= 1 - 0.7788 = 0.2212.$$

b) Let Y be the number of defective items in a box of 500, then the average number of defectives in the box is $\lambda = np = 500 \times \frac{1}{400} = 1.25$,

Hence $Y \sim Po(1.25)$

The required probability is given by

$$P(Y=0) = e^{-\lambda} = e^{-1.25} = 0.2865$$

**Example 10.4.6.** A shop sells AC machines at the rate of 2.5 pieces per week, find the probability that in a 2-week period the shop sells at least 7 AC machines.

**Solution.** Let X be the number of ACs sold in a 2-week period, then according to question $X \sim Po(2 \times 2.5)$ or $X \sim Po(5)$.

We know, $p(x) = \frac{e^{-\lambda}\lambda^x}{x!}$; $x = 0, 1, 2 \ldots\ldots\infty$

Then, $p(x \geq 7) = 1 - p(x \leq 6)$

$$= 1 - 0.7622 \text{ (using Poisson distribution table)}$$

$$= 0.2378$$

## 10.5. Normal Distribution

So far we have discussed two discrete probability distributions. Now, we shall discuss a continuous probability distribution, which is known as normal distribution. It is a very important distribution in statistics, The normal distribution was discovered by De-Moivre as the limiting case of binomial distribution in 1733. The other eminent mathematicians were Laplace and Gauss who played important role in its development. In honour of the important contribution of Gauss, the normal distribution is often called the Gaussian distribution. The distribution of heights, weights, and errors made in measuring certain physical quantities, I.Q's are just a

Similarly, the best-fitted regression line of x on y can be written in three different ways as

(i) $\hat{x} = c + dy$ ; by finding the values of c and d by the least spares method.

(ii) $\hat{x} = \bar{x} + d(y - \bar{y})$ ; when the values of d , $\bar{y}$ and $\bar{x}$ are available.

(iii) $\hat{x} = \bar{x} + r\frac{s_x}{s_y}(y - \bar{y})$ ; when the values of r, $s_x$ and $s_y$ are available.

**Remarks.** It is noted that the numerator of r, b and d are same. They have the same sign when they are calculated from the same set of data. That is if b is negative, then d and r are also negative. If r is positive, then b and d are also positive.

## 12.4. Relationship Between Correlation Coefficient and Regression Coefficients

There is a relationship between the correlation coefficient and the regression coefficients. Actually correlation coefficient is the geometric mean of the regression coefficients.

**Theorem 12.1. Correlation coefficient is the geometric mean of the regression coefficients.**

**Proof.** Suppose r is the correlation coefficient between x and y, b is the regression coefficient of y on x and d is the regression coefficient of x on y, then we have to prove

$$r = \sqrt{c \times d}.$$

The coefficient of correlation between x and y

$$r = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\sqrt{(\Sigma x-\bar{x})^2(y-\bar{y})^2}}$$

The regression coefficients of y on x and x on y are

$$b = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}; \quad d = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2}$$

$$\text{Then } b \times d = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2} \times \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2} = \left[\frac{\{\Sigma(x-\bar{x})(y-\bar{y})\}}{\sqrt{\Sigma(x-\bar{x})^2\Sigma(y-\bar{y})^2}}\right]^2 = r^2$$

Hence $r = \sqrt{b \times d}$



**Theorem 12.2. Regression coefficient is independent of the shift of origin but depends on the change of scale.**

**Proof.** Suppose $b_{y/x}$ is the regression coefficient of y on x. That is

$$b_{y/x} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}.$$

Let us define two new variables u and v by shifting the origin and changing the scale as

$$u = \frac{x-A}{h} \quad \Rightarrow x = A + hu$$

$$v = \frac{y-B}{k} \quad \Rightarrow y = B + kv$$

The regression coefficient of v on u is $b_{v/u} = \frac{\Sigma(u-\bar{u})(v-\bar{v})}{\Sigma(u-\bar{u})^2}$

Now, $b_{y/x} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}$

$$= \frac{\Sigma(A+hu-A-h\bar{u})(B+kv-B-k\bar{v})}{\Sigma(A+hu-A-h\bar{u})^2} = \frac{k\Sigma(u-\bar{u})(v-\bar{v})}{h\Sigma(u-\bar{u})^2} = \frac{k}{h}b_{v/u}.$$

This shows that regression coefficient depends on h and k but independent of A and B. This means regression coefficient is independent of the shift of origin but depends on change of scale. That is, the value of regression coefficient will remain same if we subtract a constant quantity from all the values of x and another constant quantity from the entire values y. But regression coefficient will be different if we divide all the values of x and y by two constant quantities.

## 12.5. Some Important Properties of Regression Coefficient

(i) The regression co-efficient measures the average change in dependent variable for a unit change in independent variable.

(ii) Regression coefficients are not symmetrical function of x and y. Suppose $b_{y/x}$ and $b_{x/y}$ are regression coefficients of y on x and x on y respectively, then $b_{y/x} \neq b_{x/y}$.

(iii) Both the regression coefficients have the same sign.

(iv) The correlation coefficient is the geometric mean of two regression coefficients, that is $r_{xy} = \sqrt{b_{x/y} \cdot b_{y/x}}$.